Endogenizing Nature in Production Theory: A Framework of Economic Commensalism

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Abstract

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Keywords: economic commensalism; environmental economics; rational incentives; commons; carbon markets.

1. Introduction

From an intertemporal social-efficiency perspective, informed by Pareto and Kaldor–Hicks criteria, neither extreme predation, marked by the indiscriminate and often illegal depletion of natural resources, nor strict preservationism, characterized by minimal or non-extractive use of the environment, can provide a stable and efficient resolution of the trade-off between production and ecological integrity (DASGUPTA, 2001). These two poles represent opposing boundaries of environmental interaction: the former disregards sustainability entirely, while the latter may overly constrain productive potential under conventional productivity metrics, which typically emphasize output expansion over ecological feedbacks. Situated between these extremes, we analytically describe resource

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extraction as the use of natural resources at rates that do not exceed regenerative thresholds, a condition often assumed, though rarely guaranteed, in both economic models and environmental policy. In contrast, in the political economy literature, "extractivism" denotes a development model structurally oriented toward export-driven resource depletion, one that systematically operates beyond those thresholds despite formal commitments to sustainability (GUDYNAS et al., 2009; ACOSTA, 2013; MCKAY, 2017). This paper engages with both conceptions, treating the gap between them not as a mere terminological ambiguity but as a substantive indicator of the conditions under which resource use transitions from individually rational, under prevailing price systems, property-rights arrangements, and policy regimes, to ecologically predatory, as reflected in declining future productivity, ecological threshold violations, private–social payoff divergence, and intertemporal welfare losses.

A related difficulty emerges in the way production theory represents nature. A traditional strand of economic analysis defines the factors of production as land or natural resources (N), capital (K), and labor (L), the fundamental inputs in the production of goods and services. Although ecological conditions are directly or indirectly embedded in all three, dominant strands of economic modeling have often treated the environment either as an exogenous constraint, an externality, a fixed natural endowment, or a corrective term, rather than as an endogenous and dynamic component of production itself (JEHLE; RENY, 2011). As a result, ecological constraints are frequently ignored or only indirectly considered through land productivity, which affects land rents according to natural attributes such as fertility, location, and regenerative capacity (HUBACEK; BERGH, 2002).

In response to the ongoing environmental degradation, over the past decades, a range of institutional efforts has been undertaken to curb environmental disasters such as global warming, tropical deforestation, and the indiscriminate or illegal exploitation of water and soil resources. International agreements such as the Kyoto Protocol (1997), the Paris Agreement (2015), and the Convention on Biological Diversity (1992), as well as regional initiatives and national policies, have established frameworks for emission reduction, biodiversity conservation, and sustainable resource management. Not necessarily considering human self-interest over ecological needs, these instruments often rely on legally binding commitments, emission trading systems, reforestation programs, and technological cooperation mechanisms (EASTERLY, 2013).

In a related vein, legal and institutional instruments have been established to create a system of checks and balances capable of inhibiting irrational environmental exploitation in the short and medium term, while fostering ecological awareness in the long term. For

example, the Brazilian Forest Code (Lei nº 12.651/2012) stands out as a key legal framework regulating land use and native vegetation preservation in rural properties within the country's borders (BRASIL, 2012), while non-governmental organizations (NGOs) such as Greenpeace, the World Wide Fund for Nature (WWF), the Rainforest Foundation, Imazon (Instituto do Homem e Meio Ambiente da Amazônia), among many others, play a crucial role in monitoring compliance, raising public awareness, and pressuring both governments and corporations to adopt more sustainable practices throughout the world (LOMBORG, 2001).

Despite their relevance, such efforts often reflect a moralistic or idealized form of environmentalism. While rhetorically aimed at fostering mutualistic relationships between humans and nature, they often result in a regulated form of extractivism. Legal frameworks such as the Forest Code impose rigid constraints that, while intended to protect native ecosystems, fail to align with the economic rationale of landowners, who often perceive them not as incentives for sustainable coexistence, but as punitive restrictions that prevent them from fully utilizing a productive factor — land — under their control (AGRAWAL, 2005). This maladjustment not only generates resistance and noncompliance but also fuels a broader sense of injustice among rural producers, who feel disproportionately burdened by conservation obligations without corresponding compensatory mechanisms (BöRNER et al., 2016).

Theoretical and applied economics, in turn, has addressed environmental issues with the common understanding that markets fail to internalize ecological costs. Without some form of coercive intervention — usually by governments — such failures tend to generate vicious long-term outcomes. For instance, Pigou (2010) have proposed tariffs (Pigouvians taxes) as a way to internalize negative environmental externalities alongside its neoclassical model. In his view, Hardin (1968) argued that, in the absence of private property or state regulation, individual self-interest leads to what he called the "tragedy of the commons", a situation in which natural resources are completely exhausted as a result of individuals aiming to maximize their own utility, while the common good is entirely neglected.

Alternatively, Ostrom (2002) challenged Hardin's dilemma by arguing that, under certain institutional conditions, the self-management of natural resources can be sustainable. Assunção et al. (2013) highlight that enforcement, when paired with well-designed incentives, has shown results in reducing Amazon deforestation, yet faces structural and political resistance when scaled. Scheinkman (2008)'s contributions to the modeling of local interactions and externalities provide a relevant foundation for understanding how decentralized behavior can lead to socially suboptimal equilibria, especially in resource

use contexts.

Since the mid-1980s, the so-called Green Economy has advanced the argument that economic development is only viable if improvements in human well-being are balanced with reductions in environmental risks and ecological scarcity (JACOBS, 1991). According to this school of thought, **the economy must stop treating the environment as a substitute good and begin to value it as a complementary asset**. This implies ensuring that each generation inherits an ecosystem that is at least equivalent in quality and function, recognizing that many ecological characteristics are irreplaceable. Such an approach is framed not only by the risks of irreversibility and uncertainty but also by the moral imperative of intergenerational equity (PEARCE et al., 1989).

As we can see, throughout history, a persistent misalignment has existed between collective supply and private demand in the use of natural resources, which leads to misconduct on the part of certain private agents who do not consider it fair to internalize diffuse social costs in individual operations, all of this intertwined in a non-standardized and inefficient institutional-legal scheme, operated by States and monitored by the third sector. Despite advances in theory and policy, a deeper problem persists: the economic system — including markets, institutions, and even economic theory — struggles to adequately conceptualize and resolve the trade-off between development and sustainability.

Apparently, an endless back-and-forth dilemma is presented as an unsolvable tension between collective morality and individual utilitarianism. In the end, nature seems to be on a path of degradation that initially affects vulnerable regions, but eventually reaches everyone, as environmental collapse leads to widespread economic consequences such as global warming, water scarcity, and the increasing threat to agricultural and ecological food systems. This apparent antagonism may in fact be rooted in theoretical and institutional misalignments, rather than from an inherent incompatibility between individual incentives and ecological responsibility, a conjecture explored throughout this study.

1.1. Motivations

As demonstrated, despite their normative relevance and global reach, the implementation and effectiveness of actions taken by governments, NGOs, the media, academia, and public opinion are persistently challenged by the economic interests and political imperatives of productive agents and decision-makers. In many cases, short-term economic gains and competitive pressures continue to override long-term sustainability goals, limiting the transformative potential of these institutional frameworks.

The practical outcome of this trade-off between economics and ecology has often been the over-exploitation of natural resources, leading to negative externalities such as

the depletion of land productivity, pollution of air, soil, rivers, and other water sources, disruption of hydrological cycles, increase in agricultural pests, and rising production costs. These impacts have not only degraded biomes and ecosystems but also compromised the quality of life and the productive capacity of the economic agents involved. The indiscriminate use of natural resources — treated as exogenous and not subject to scarcity analysis — has led to reactive approaches aimed at correcting the resulting instabilities, such as the overuse and declining utility of key productive inputs like land, water, and air.

Mato Grosso serves as the heart of Brazil's agricultural expansion and hosts three primary biomes: the Amazon, Cerrado, and Pantanal. It offers a compelling empirical and conceptual context for this study. The region faces escalating challenges from deforestation, rising global commodity needs, and new sustainability obligations. These factors position the state at the intersection of risks and opportunities in the move towards environmentally sustainable practices that align with economic incentives. The region's blend of significant environmental value and robust production creates a scenario to test if rational eco-economic dynamics — similar to those in the economic commensalism framework — can harmonize productivity with sustainable conservation without relying solely on coercion or moral appeals.

Typical examples of this concern, a dual perspective emerges from the land and burgeoning carbon markets, shedding light on the complexities of environmental and economic interactions. Land is still regarded as a static input within traditional production models, yet its ecological value, particularly when conserved, tends to lack proper valuation or is grossly undervalued. Likewise, carbon, although recognized as a marketable commodity in both voluntary and regulatory contexts, often fails to reflect the spatial dynamics of land utilization and ecosystem sturdiness. These discrepancies create contradictory motivations for producers, who gain incentives to enlarge their productive areas but face indirect penalties for land conservation, despite the potential for conservation to offer quantifiable ecological benefits and financial gains. This discord highlights the necessity for a novel theoretical framework that integrates the environment as a core economic element.

Given the opportunity costs faced by producers and consumers, traditional solutions — such as Pigouvian taxes, regulatory enforcement, community governance, and incentive-based schemes — have provided only partial answers. These approaches often fail to account for long-term ecological uncertainty, interdependent decision-making among producers, and the endogenous learning behaviors — involving agricultural producers, ecologists, politicians, consumers, public opinion and legal instruments — that

shape conservation outcomes. This conceptual and institutional gap motivates the search for a more integrated and incentive-compatible framework, which constitutes the foundation of the present study.

1.2. Rationale

Despite decades of theoretical development, current approaches to environmental economics still fail to endogenize natural resources within production decisions. Orthodox models treat the environment as an exogenous factor, while corrective instruments such as taxes or command-and-control mechanisms face limitations in efficiency, grounded in assumptions such as the environment being regulatable as a nearly homogeneous entity, carbon being a universal proxy for compensation, and human rationality being static and ubiquitous.

Thus, although significant theoretical advances and practical attempts have been made in integrating environmental concerns into economic decision-making, important shortcomings remain. Taxes, legal embargoes, property rights, and — as the following chapters will demonstrate through theoretical and empirical evidence — carbon markets have fallen short of delivering on both environmental and economic expectations.

1.3. Justification

Although carbon markets, environmental certifications, regulatory constraints, and coercive enforcement mechanisms have become widespread, these tools have not been effective in changing the core economic behaviors that lead to environmental harm. Carbon credits frequently lack connection to the specific ecological settings and fail to provide incentives that are accessible, predictable, or appealing to producers. Certifications primarily advantage a limited group already aligned with sustainability ideals, while legal enforcement struggles with weak institutions, limited monitoring capabilities, and strategic circumvention. These measures often function outside the decision-making frameworks of agents, viewing the environment as a limitation rather than an opportunity. This disconnect suggests the necessity for a model where environmental preservation is embedded within agents' rational decision-making, not as an external requirement, but as a strategic advantage, which is the foundation of the economic commensalism framework presented in this study.

These limitations reveal the need for a new paradigm capable of reconciling production incentives with environmental preservation. In this context, the concept of *economic commensalism* is proposed as a stable framework through which economic agents

can align individual profitability with collective ecological sustainability, provided that suitable incentive structures are in place.

This work aims to be relevant by highlighting how economic theory can be improved, and this will not only clarify what and why current environmental preservation mechanisms do not work properly — which include the carbon market, environmental certificates and targets for controlling the use of natural resources —, but will also provide, through a stable and well-founded theoretical framework, ways to improve market instruments and state actions related to environmental preservation without necessarily compromising the economic interests of the agents involved.

2. World and Brazilian Evidence

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